Remote Workforce & Employee Well-being Analytics

Business Scenario

A multinational company has adopted remote and hybrid work policies across multiple regions. HR leadership wants to understand how remote work affects employee productivity, stress levels, work-life balance, mental health, and employee satisfaction.

As a **Power BI Data Analyst**, your objective is to transform raw employee data into an interactive executive dashboard that helps HR leaders identify risks, improve employee well-being, and make data-driven workforce decisions.

Dataset

Column	Description
Employee_ID	Unique Employee ID
Age	Employee Age
Gender	Gender
Job_Role	Job Role
Industry	Industry
Years_of_Experience	Experience
Work_Location	Remote / Hybrid / On-site
Hours_Worked_Per_Week	Weekly Working Hours
Number_of_Virtual_Meetings	Weekly Virtual Meetings
Work_Life_Balance_Rating	Rating (1-5)
Stress_Level	Low / Medium / High
Mental_Health_Condition	Mental Health Status
Access_to_Mental_Health_Resources	Yes / No
Productivity_Change	Increased / No Change / Decreased
Social_Isolation_Rating	Rating

Column	Description
Satisfaction_with_Remote_Work	Rating
Company_Support_for_Remote_Work	Rating
Physical_Activity	Hours/Frequency
Sleep_Quality	Rating
Region	Employee Region

Business Objectives

The HR Director wants to answer the following:

- Is remote work improving productivity?
- Which employees are experiencing the highest stress?
- Which regions require better HR support?
- Does company support improve employee satisfaction?
- How are work-life balance and productivity related?
- Which job roles are at the highest risk of burnout?

Project Deliverables

Create:

- Interactive Power BI Dashboard
- KPI Cards
- Executive Summary
- Drill-through Pages
- Dynamic Tooltips
- Slicers
- DAX Measures
- Power Query Transformations

Phase 1 – Data Preparation (Power Query)

Tasks

Task 1

Import the dataset.

Task 2

Verify data types.

Task 3

Check for duplicate Employee IDs.

Task 4

Handle missing values.

Task 5

Rename columns for readability.

Task 6

Create Age Groups:

- Under 25
- 25–34
- 35–44
- 45–54
- 55+

Task 7

Create Experience Groups:

- Fresher
- Junior
- Mid-Level
- Senior
- Expert

Task 8

Create Working Hours Category:

- Less than 35
- 35–45
- 46–55
- More than 55

Task 9

Standardize text values (trim, proper case where appropriate).

Phase 2 – Data Modelling

Create a Star Schema (Single Table Scenario)

Although the dataset is a single table, organize it using logical dimensions:

- Employee Attributes
- Work Characteristics
- Mental Health Indicators
- Productivity Indicators

Create a Date Table only if future enhancements include time-based data.

Phase 3 – DAX Calculations

Basic Measures

- Total Employees
- Average Age
- Average Experience
- Average Working Hours
- Average Virtual Meetings
- Average Work-Life Balance
- Average Satisfaction Rating
- Average Sleep Quality
- Average Social Isolation
- Average Company Support Rating

Employee Distribution Measures

- Employees by Gender
- Employees by Work Location
- Employees by Region
- Employees by Job Role
- Employees by Industry

Productivity Measures

- Employees with Increased Productivity
- Employees with No Productivity Change
- Employees with Decreased Productivity
- Productivity Improvement Percentage

Mental Health Measures

- High Stress Employees
- Medium Stress Employees
- Low Stress Employees
- Employees with Mental Health Conditions
- Employees with Access to Mental Health Resources
- Access Percentage

Satisfaction Measures

- Average Satisfaction
- Average Work-Life Balance
- Average Company Support
- Average Sleep Quality
- Average Physical Activity

Phase 4 – Dashboard 1: Executive Overview

KPI Cards

- Total Employees
- Average Work-Life Balance
- Average Stress Level (mapped numerically)
- Average Satisfaction
- Average Sleep Quality
- Average Productivity Score (mapped numerically)

Visuals

- Employees by Region (Map or Filled Map if region supports it)
- Employees by Work Location (Donut Chart)
- Employees by Gender (Pie/Donut Chart)
- Employees by Industry (Bar Chart)
- Employees by Job Role (Bar Chart)

Dashboard 2 – Productivity Analytics

Business Questions

- Which work location has the highest productivity?
- Does experience affect productivity?
- Does the number of meetings impact productivity?

Suggested Visuals

- Productivity Change by Work Location
- Productivity Change by Industry
- Productivity by Job Role
- Productivity by Experience Group
- Productivity by Hours Worked
- Productivity by Virtual Meetings

Dashboard 3 – Employee Well-being

Business Questions

- Which employees report the highest stress?
- Does work-life balance reduce stress?
- Does sleep quality impact stress?
- Which regions experience poor work-life balance?

Suggested Visuals

- Stress Level Distribution
- Stress by Region
- Stress by Job Role
- Work-Life Balance by Work Location
- Sleep Quality by Stress Level
- Physical Activity by Stress Level

Dashboard 4 – Mental Health Analytics

Business Questions

- How many employees have access to mental health resources?
- Does access improve satisfaction?
- Does access reduce stress?

Suggested Visuals

- Access to Resources (Donut Chart)
- Satisfaction by Resource Access
- Stress by Resource Access
- Mental Health Condition by Region
- Mental Health Condition by Job Role

Dashboard 5 – Remote Work Analysis

Business Questions

- Which work location provides the best work-life balance?
- Which work location has the happiest employees?
- Which work location receives the highest company support?

Suggested Visuals

- Satisfaction by Work Location
- Company Support by Work Location
- Work-Life Balance by Work Location
- Productivity by Work Location
- Stress by Work Location

Dashboard 6 – HR Insights

Create an executive page answering:

Which regions have:

- Highest stress?
- Lowest satisfaction?
- Highest productivity?
- Lowest company support?

Which job roles have:

- Highest stress?
- Lowest work-life balance?
- Highest social isolation?
- Lowest sleep quality?

Drill-through Pages

Create drill-through pages for:

- Region

- Job Role
- Industry

Each page should display:

- Employee count
- Average stress
- Average work-life balance
- Average satisfaction
- Average productivity
- Average sleep quality
- Average social isolation

Report Interactivity

Use:

- Slicers: Region, Industry, Work Location, Gender, Job Role, Stress Level
- Drill-through
- Cross-filtering
- Dynamic Titles
- Dynamic Tooltips
- Bookmarks for page navigation

- Hierarchies (Region → Industry → Job Role)

Capstone Business Questions

1. What percentage of employees work remotely, hybrid, and on-site?
2. Which work location has the highest employee satisfaction?
3. Which job roles experience the highest stress levels?
4. Does higher work-life balance correlate with increased productivity?
5. Which regions have the greatest social isolation?

6. Does access to mental health resources improve employee satisfaction?
7. How do working hours affect stress and work-life balance?
8. Does sleep quality appear to influence productivity?
9. Which industries report the highest stress?
10. Which experience group has the highest satisfaction?
11. Which work location receives the strongest company support?
12. Which regions should HR prioritize for wellness initiatives?

Expected Dashboard Pages

Page	Focus
Executive Overview	Workforce summary and KPIs
Productivity Analytics	Productivity drivers and trends
Employee Well-being	Stress, work-life balance, sleep, physical activity
Mental Health	Conditions, resource access, and support
Remote Work Analysis	Comparison of Remote, Hybrid, and On-site employees
HR Insights	Actionable recommendations by region, industry, and job role